WASP-10b

R-1670

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_140924` · created 2026-08-02T10:27:30+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456924.8228 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.0058
Transit depth (Rp/Rs)^2	1.99 +/- 0.16 [%]
Orbital Inclination (inc)	89.81 +/- 0.52
Airmass coefficient 1 (a1)	0.989 +/- 0.0071
Airmass coefficient 2 (a2)	0.01 +/- 0.0089
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	#2 - [201, 344]
Optimal Aperture	3.5
Optimal Annulus	10.18
Transit Duration (day)	0.09331 +/- 0.00079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.0058 vs 0.1592 (deviation 3.02σ)
Duration fit vs published	0.09331 d vs 0.0946 d (deviation 1.55σ)
Mid-time O–C	0.6 min
Depth SNR	23.4
Residual scatter	0.71 %

Light curve

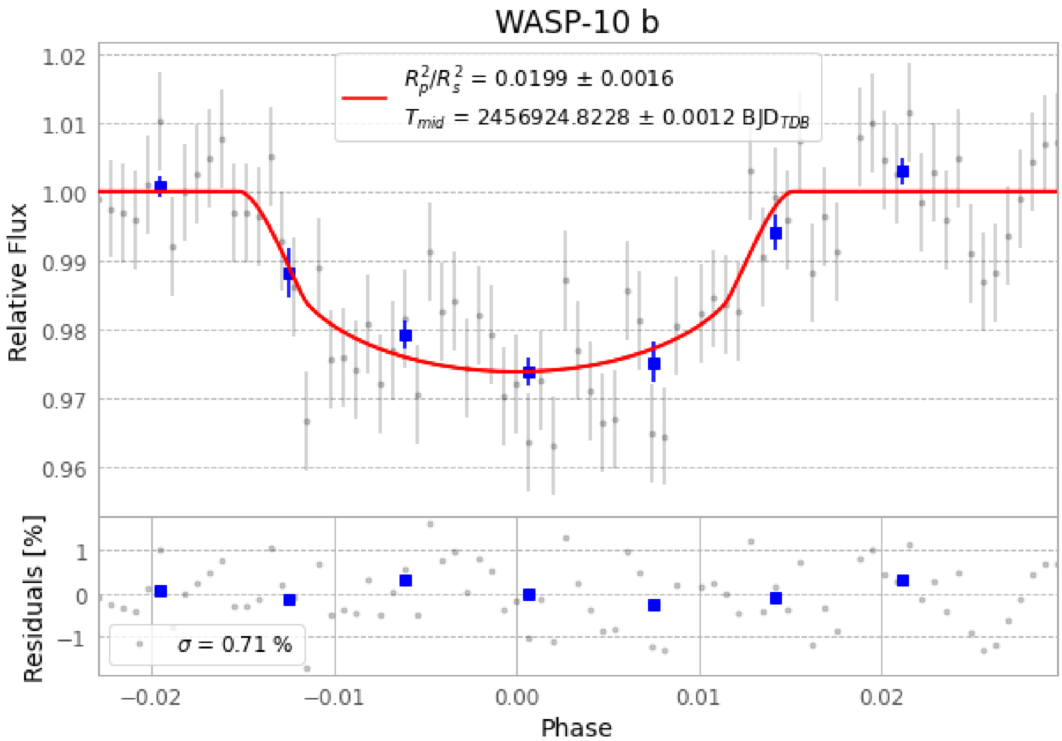

Figure 1 — FinalLightCurve_WASP-10 b_2014-09-23.png (SHA-256 9ecf64c9d4a9aa1b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [236, 272], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 374efe09dc5c3136…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

79 FITS frames (52 MB), `WASP-10140924055412.FITS` → `WASP-10140924094813.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-140924.log (54a415ae0f5d…), FinalLightCurve_WASP-10 b_2014-09-23.png (9ecf64c9d4a9…), FinalLightCurve_WASP-10 b_2014-09-23.csv (90afff25cdb3…)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 10:27Z.